

P1 Open Path - Quartic Oscillator

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This note covers some result implemented in the code for quartic oscillator. We will use the convention of $m = \hbar = 1$

The momentum distribution is given by the Fourier transform of the end to end distribution $N(\Delta)$

$$n(p) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} d\Delta e^{ip\Delta} N(\Delta) \quad (1)$$

$$N(\Delta) = \frac{\langle e^{-(\Delta-R)^2/(2\beta_P)} \rangle_o}{\langle e^{-R^2/(2\beta_P)} \rangle_o}. \quad (2)$$

where $R = q_P - q_1$, $\beta_P = \beta/P$, $\omega_P = 1/\beta_P$, $\mathbf{p} = \{p_k\}_{k=1}^P$, $\mathbf{q} = \{q_k\}_{k=1}^P$ and

$$\langle \dots \rangle_o = \frac{\int d\mathbf{p} \int d\mathbf{q} e^{-\beta_P H_P^{(o)}(\mathbf{p}, \mathbf{q})} (\dots)}{\int d\mathbf{p} \int d\mathbf{q} e^{-\beta_P H_P^{(o)}(\mathbf{p}, \mathbf{q})}} \quad (3)$$

where

$$H_P^{(o)}(\mathbf{p}, \mathbf{q}) = \sum_{k=1}^P \left[\frac{p_k^2}{2} + \frac{q_k^4}{4} \right] + \frac{\omega_P^2}{2} \sum_{k=1}^{P-1} (q_k - q_{k+1})^2. \quad (4)$$

one can easily check $n(p)$ is normalised by integrating over p , and see the factor of $1/2\pi$ is required for the Dirac delta factor and $N(0) = 1$. Substitute Eq.(2) back into Eq.(1), then letting $u = \Delta - R$ and doing the integral gives

$$n(p) = \sqrt{\frac{\beta_P}{2\pi}} e^{-\beta_P p^2/2} \frac{\langle e^{ipR} \rangle_o}{\langle e^{-R^2/(2\beta_P)} \rangle_o}. \quad (5)$$

Notice that reversing the label of the beads from $1, 2, 3, \dots, P \rightarrow P, P-1, \dots, 1$ does not change the Boltzmann weight $e^{-\beta_P H_P^{(o)}(\mathbf{p}, \mathbf{q})}$, but changes $R \rightarrow -R$, thus by symmetry

$$\langle \sin pR \rangle_o = 0 \quad (6)$$

and we end up with a rather simple equation to implement

$$n(p) = \sqrt{\frac{\beta_P}{2\pi}} e^{-\beta_P p^2/2} \frac{\langle \cos pR \rangle_o}{\langle e^{-R^2/(2\beta_P)} \rangle_o}. \quad (7)$$